

PluralTree: What Is and Is Not Established

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Companion to `methodology.tex`, which defines the method. This document states what the measurements support, at what strength, and what remains untested. Prose results as they land are recorded in `findings_2026_09_15.md`; this file is the version that belongs beside the method.

1 Instrument

All coverage numbers come from an LLM judge whose own validation is unresolved. Its aggregate agreement with human ratings is $\rho = 0.167$, against a ceiling of $\sqrt{0.574} \approx 0.76$ set by the benchmark’s own human split-half reliability (0.574 after the Spearman–Brown correction). With only eight reference models the sampling error at that level is very wide (split-half range $[-0.17, +0.83]$), so the ranking-level check can neither validate nor invalidate a judge. Read every item below through that.

2 Established

- `MERGE_v2` > baseline replicates on the reference embedding: $+0.039$, $+0.050$ ($p=0.014$), $+0.048$ ($p=0.024$) across independent generations at 72B. On freshly trained embeddings it is marginal in both geometry arms ($+0.040$, $p=0.062$; $+0.045$, $p=0.061$; three rollouts), so embedding-training variance may matter.
- The gain transfers to a second, independently built survey graph. On one ISSP module, restricted to the 34 of 60 questions whose anchors resolve, `MERGE_v2` gains $+0.048$ (95% CI $[+0.005, +0.094]$, $p=0.031$); on the 26 unresolved questions, where the arm runs without forks, the difference is -0.008 ($p=0.65$). Pooled over all 60 the effect dilutes to $+0.024$ ($p=0.11$), as attenuation predicts.
- The retrieval instrument is valid. Within an axis — the only comparison fork selection makes — the learned hyperbolic distance ranks the Jensen–Shannon divergence between subgroups’ answer distributions at $\rho = +0.755$ (2,027 axes; pooled $+0.529$; shuffled null -0.000), while the encoder’s own input features rank it at only $+0.074$. The encoder recovers ordering its inputs do not expose.
- The merge is faithful: median retention 1.0, with the semantic guard firing on 1 of 60 merges. Where coverage is lost at 7B, the loss is in draft quality, not compression.
- CAD at 7B is monotone in α across five values (0.352, 0.327, 0.099, 0.047, 0.042 for $\alpha = -0.5..+0.5$): the strength of context influence controls coverage, and suppressing it helps.

3 Ruled out as drivers of coverage

- *Curvature.* At three rollouts the hyperbolic and Euclidean arms converge (+0.040 vs +0.045; $\Delta\Delta = +0.005$, 95% CI $[-0.040, +0.057]$, $p=0.88$), a well-powered null, even though the hyperbolic encoder fits the tree markedly better (tree-distance ρ 0.64 vs 0.29; sibling ratio 0.65 vs 0.08). An earlier one-rollout run appeared to separate the arms ($\Delta\Delta$ at $p=0.094$); it did not replicate.
- *Hierarchy.* Flat retrieval — no anchor, no descent, nearest opinion leaves paired into pseudo-forks — ties MERGE_v2 (+0.010 in flat’s favour, 95% CI $[-0.011, +0.031]$, $p=0.34$), with fork volume matched (5.00 vs 4.85) and lower divergence (11.9 vs 13.7). Any hierarchy benefit is bounded near +0.01.
- *Per-question routing.* On 60 questions no feature set beats the trivial always-inject policy, which recovers +0.525 of the oracle gap; graph features recover -0.040 and the full causal set -0.178 , below length alone (+0.118). Five earlier routers failed the same way.

4 Not established

- *That divergence selection matters.* Random-pair (DIVRAND) and distribution-ranked (JSMAX) selection arms have not reported. The disagreement being ranked is small in absolute terms: mean JS = 0.0154 bits over same-axis subgroup pairs, so the most divergent available pair is often barely divergent.
- *That fork content is load-bearing.* The matched irrelevant-fork control (RAND) was confounded in every prior run.
- *That the gain is not length.* Injected answers run roughly five times longer than baseline’s and no length-matched baseline has been run.
- *The reward.* Reward–judge concordance within question is 0.15 against chance 0.5; the reward scores survey positions, the judge scores participant clusters, and no threshold reconciles them. The group-diversity reward replacing it is gated on the same statistic and has not yet been measured.
- *CAD’s magnitude.* The no-forks reference (0.394) uses a different instruction from the CAD arms, so $0.394 \rightarrow 0.099$ conflates the fork block with the instruction. The $\alpha = -1$ arm would separate them and has not been run.
- *The 7B collapse’s size.* At 7B MERGE_v2 falls 0.17–0.26 below baseline, but measured with a 7B judge; the 72B re-judge is pending.

5 Known trade-off

Across-sample diversity (INFINITY-CHAT): MERGE_v2 is *less* diverse than baseline on every metric, losing 84–89 of 100 queries ($p \approx 0$). It trades across-sample diversity for within-answer coverage. Retrieval resolved on 71 of the 100 queries.

6 Where coverage is lost

Of the viewpoint clusters the benchmark scores, 29.7% (ATP) and 45.4% (ISSP) are covered by no arm at all, and those clusters are smaller than average (relative size 0.278 on ISSP). The clusters an arm *does* recover beyond baseline are larger than a typical miss (0.53 vs 0.45), so no arm is reaching the minority end the benchmark exists to reward. Which stage loses them — retrieval, drafting, merging or judging — is the question `scripts/analysis/coverage_funnel.py` was built to answer.